

Human vs GPT-4o-mini Cloze Predictions

1980-01-01

Task and Data

A **cloze task** presents a sentence with the final word removed; respondents supply what comes next. The dataset covers **144 Russian sentence contexts**. Human responses were collected from 628 subjects (14–151 per context, avg 48), yielding 6 898 individual responses with 1–65 unique answers per context. GPT-4o-mini predictions were obtained as a ranked list of token continuations with log-probabilities, converted to probabilities via $\exp(\sum \text{logprob})$. After deduplication by stripped surface form, each context has between 6 and 13 957 distinct GPT candidates.

Metrics

Human accuracy — the proportion of human respondents who gave the exact ground-truth target word. In 62 % of contexts (89/144) no human guessed the target at all; mean accuracy across all contexts is 0.10 (median 0.00), reaching 1.00 only for the easiest items.

match@k — for a given k , the fraction of **unique** human answer types that appear anywhere in GPT’s top- k predictions (unweighted).

weighted match@k — the fraction of **human response mass** covered: for each human answer found in the top- k , its probability (response frequency) is summed. Because popular answers contribute more, weighted match is consistently higher than unweighted match and is the more meaningful metric.

Results

Coverage saturates rapidly: almost all gains occur before $k = 10$, with negligible improvement beyond $k = 20$.

k	match@ k	weighted match@ k
1	0.064	0.180
5	0.176	0.363
10	0.208	0.398
20	0.213	0.404
100	0.216	0.406
1000	0.216	0.406

At $k = 10$ the model’s top candidates already cover 40 % of human response mass on average, but only 21 % of unique answer **types** — reflecting that GPT reliably finds the most frequent human answers while missing rare or idiosyncratic ones.

In 9 contexts (e.g. *летела, миссионеры, обгорели*) GPT produces **zero overlap** with any human answer even at $k = 1000$, suggesting fundamental distributional divergence on semantically constrained or low-frequency items.

Summary

GPT-4o-mini’s probability mass concentrates on the same words humans prefer most, but its vocabulary of plausible continuations is much narrower: the long tail of human creativity (typos

aside) is largely absent from the model's distribution. The sharp saturation of $\text{match}@k$ by $k = 10$ means that simply taking the model's top-10 candidates is nearly as informative as its entire ranked list.